Artificial Intelligence-Driven Malware Detection via Windows Event Log Engineering in Smart Grid Control Systems

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Agenda

Motivation

Related Work

Threat Model

Data and Features

Model

Results

Discussion

Motivation

Smart Grid Cyber Threat Landscape

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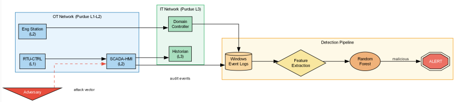
- High-profile incidents: Stuxnet [1], Ukrainian grid attacks (2015, 2016), Colonial Pipeline (2021)
- Regulatory mandates: NERC CIP-007 requires event logging; IEC 62351 specifies audit trails
- **Gap**: Windows Event Logs at the OT layer are untapped for intrusion detection



Related Work

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- **Drain** [2] — online log parsing with fixed-depth tree
- **DeepLog** [3] — LSTM sequence modelling of log keys
- **LogAnomaly** [4] — semantic vector anomaly detection

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- **LogAnomaly** [4] — semantic vector anomaly detection
- **Common limitation**: none applied to ICS/SCADA Windows hosts or ICS-specific event semantics

Related Work: ICS Intrusion Detection

- **Carcano et al.** [5] — critical-state analysis for SCADA command sequences
- **Goldenberg & Wool** [6] — Modbus/TCP protocol modelling
- **SWaT dataset** [7] — physical/network measurements, no host audit logs

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- **Mitchell & Chen** [8]: host-level IDS is rare in CPS literature
- **Our contribution**: first framework targeting Windows Event Logs at Purdue L1–L3

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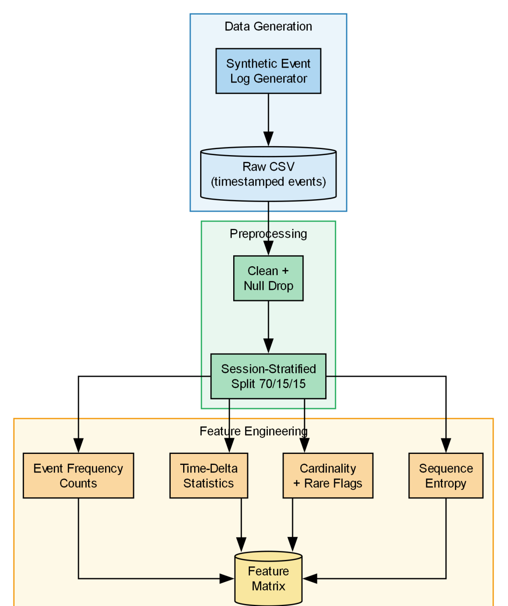
4 Attack Patterns Modelled (MITRE ATT&CK ICS)
Lateral movement · Persistence · Privilege escalation · Reconnaissance

Data and Features

Synthetic Event Log Generator

- 6,000 sessions: 5,000 benign + 1,000 malicious (5:1 ratio)
- 54,478 total events
- 13 Windows Event IDs (4624, 4625, 4663, 7045, …)



Session-Level Feature Engineering

Group	Features	Dim.
Event frequency counts	count per event ID + total	14
Time-delta statistics	mean, std, min, max gap	4
Host/user cardinality	unique sources, users, hosts	3
Rare-event flags	binary indicators	var
Sequence entropy	Shannon H over event IDs	1

Total: 24-dimensional feature vector per session

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- Sub-millisecond inference latency per session

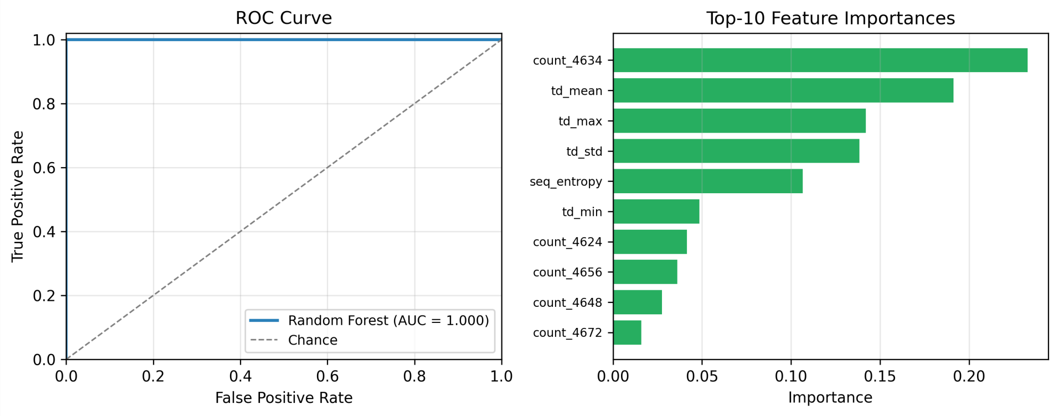
Training Protocol
RandomizedSearchCV (20 configs) · 5-fold stratified CV · macro F1 objective

Results

Classification Results

Method	Accuracy	F1 (macro)	ROC-AUC	FPR
Logistic Regression	0.952	0.943	0.980	0.048
Decision Tree	0.979	0.972	0.975	0.021
SVM (RBF)	0.967	0.961	0.986	0.033
Random Forest	**1.000**	**1.000**	**1.000**	**0.000**

ROC Curve and Feature Importances



Ablation Study

Removed Feature Group	F1 (macro)	ΔF1
None (full)	1.000	—
Group 2: Time-delta stats	0.831	-0.169
Group 5: Sequence entropy	0.947	-0.053
Group 1: Event counts	0.976	-0.024
Group 3: Cardinality	0.982	-0.018
Group 4: Rare-event flags	0.994	-0.006

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Time-delta and entropy features are the most critical.

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- **Concept drift**: periodic retraining on sliding windows; ADWIN drift detector
- **GDPR**: pseudonymise usernames/hostnames at collection; align retention with NERC CIP 90-day requirement
- **Latency**: 0.34 ms median per session on commodity hardware — SIEM-compatible

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- First open-source framework for Windows Event Log-based malware detection in smart grid ICS
- 24-feature session pipeline; Random Forest with class-imbalance compensation
- Perfect synthetic benchmark: $\text{F1} = 1.00$, $\text{ROC-AUC} = 1.00$
- Temporal features most discriminative (ablation $\Delta\text{F1} = -0.169$)
- Framework enables reproducible ICS security research without operational data

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Future Work
Real ICS logs · Multi-class ATT&CK attribution · Online learning · Adversarial robustness

[1] R. Langner, **"Stuxnet: Dissecting a cyberwarfare weapon,"** *IEEE Security & Privacy*, vol. 9, no. 3, pp. 49–51, 2011. DOI: 10.1109/MSP.2011.67

[2] P. He, J. Zhu, Z. Zheng, and M. R. Lyu, **"Drain: An online log parsing approach with fixed depth tree,"** in *2017 IEEE International Conference on Web Services (ICWS)*, 2017, pp. 33–40. DOI: 10.1109/ICWS.2017.13

[3] M. Du, F. Li, G. Zheng, and V. Srikumar, **"DeepLog: Anomaly detection and diagnosis from system logs through deep learning,"** in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017, pp. 1285–1298. DOI: 10.1145/3133956.3134015

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[7] J. Goh, S. Adepu, K. N. Junejo, and A. Mathur, **"A dataset to support research in the design of secure water treatment systems,"** in *Critical Infrastructure Protection X*, 2016, pp. 88–112. DOI: 10.1007/978-3-319-48737-3_5

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Questions

Thank you.

Questions welcome.